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Skill : WEEK 7

Task -1:

To Parse Input Redirection, Open Files, Redirectstdin, Handle MissingFiles, Restore

Streams,Test File Inputs

Code:

```
UW PICO 5.09 File: 81.c Modified
// If execvp fails
perror("execvp failed");
exit(1);
}

// Parent process
close(fd);

waitpid(pid, NULL, 0);

printf("\nCommand execution completed.\n");

return 0;
}

Get Help      WriteOut    Read File    Prev Pg      Cut Text     Cur Pos
Exit          Justify     Where is    Next Pg      UnCut Text   To Spell
```

```
Last login: Sat Oct 3 02:44:16 on ttys000
nrk@nallas-mac-Book-Air ~ % nano 81.c

nrk@nallas-mac-Book-Air ~ % gcc 81.c -o 81
nrk@nallas-mac-Book-Air ~ % ./80
zsh: no such file or directory: ./80
nrk@nallas-mac-Book-Air ~ %
```

2.To Parse OutputRedirection, Create OutputFiles, Redirect stdout,Handle Permissions, Verify Output Data, Test Redirection

Code:

```
UW PICO 5.09 File: 82.c Modified
}
    return 1;
}

printf("\n--- Output File Contents ---\n");
char buffer[256];
while (fgets(buffer, sizeof(buffer), fp) != NULL) {
    printf("%s", buffer);
}

fclose(fp);
printf("\n--- Verification Complete ---\n");
return 0;
}
```

Get Help Exit WriteOut Justify Read File Where is Prev Pg Next Pg Cut Text UnCut Text Cur Pos To Spell

```
Last login: Sat Oct 3 08:47:02 on ttys000
nrik@nallas-mac-Book-Air ~ % nano 82.c

nrik@nallas-mac-Book-Air ~ % gcc 82.c -o 82
nrik@nallas-mac-Book-Air ~ % ./82
Enter command with output redirection:
Example: ls -l > output.txt
```

3) To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios

```
LM PICO 5.09 File: 85.c Modified
}
token = strtok(NULL, " \t");
}

args[count] = NULL;

if (count == 0) {
    fprintf(stderr, "Error: Empty command.\n");
    exit(1);
}

// Execute command
execvp(args[0], args);

// Execute error
perror("execvp failed");
exit(1);
}

// Parent waits for child
waitpid(pid, NULL, 0);

printf("\nAppend operation completed.\n");

// Verify the appended data
FILE *fp = fopen(file, "r");

if (fp == NULL) {
    perror("Error reading output file");
    return 1;
}

printf("\n--- File Contents After Append ---\n");
char buffer[256];
while (fgetc(buffer, sizeof(buffer), fp) != NULL) {
    printf("%s", buffer);
}

fclose(fp);

printf("\n--- Append Validation Complete ---\n");

return 0;
}

Get Help      WriteOut      Read File      Prev Pg      Cut Text      Cur Pos
Exit          Justify       Where is       Next Pg      UNcut Text    To Spell
```

Output:

```
Last login: Sat Oct 3 08:51:55 on ttys000
nrk@nallas-mac-Book-Air ~ % nano 85.c

nrk@nallas-mac-Book-Air ~ % gcc 85.c -o 85
nrk@nallas-mac-Book-Air ~ % ./85
Enter command with append redirection:
Example: echo Hello >> output.txt
```

4) To Parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases

Code:

```
UN PICO 5.09 File: 86.c Modified

while (token != NULL && count < 99) {
    args[count++] = token;
    token = strtok(NULL, " \t");
}

args[count] = NULL;

if (count == 0) {
    fprintf(stderr, "Error: Empty command.\n");
    exit(1);
}

// Execute command
execvp(args[0], args);

// If execvp fails, error goes to redirected stderr
perror("execvp failed");
exit(1);
}

// Parent process
int status;

waitpid(pid, &status, 0);

printf("\nCommand execution completed.\n");

// Verify the error output file
FILE *fp = fopen(file, "r");

if (fp == NULL) {
    perror("Error reading error output file");
    return 1;
}

printf("\n--- Captured Error Output ---\n");
char buffer[256];

while (fgets(buffer, sizeof(buffer), fp) != NULL) {
    printf("%s", buffer);
}

fclose(fp);

printf("\n--- Verification Complete ---\n");

// Check command status
if (WIFEXITED(status)) {
    printf("Exit Status: %d\n", WEXITSTATUS(status));
} else {
    printf("Command terminated abnormally.\n");
}

return 0;
}

Get Help WriteOut Read File Prev Pg Cut Text Cur Pos
Exit Justify Where is Next Pg UnCut Text To Spell
```

Output:

```
nrk@nallas-mac-Book-Air ~ % nano 86.c
nrk@nallas-mac-Book-Air ~ % gcc 86.c -o 86
nrk@nallas-mac-Book-Air ~ % ./86
Enter command with stderr redirection:
Example: ls nonexistent 2> error.txt
1
Error: stderr redirection operator '2>' not found.
nrk@nallas-mac-Book-Air ~ %
```